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Groupe (3)

Sheet (4)

1- System modeling is the Process of developing abstract models of a system, with each model Presenting a different view or prespective of that system, it helps analysts to understand the functionality of the system & models are used to communicate with customers.

2- Models of the existing system are used during requirements engineering. They help clarify what the existing system does & Can be used as a basis for discussing its strengths & weakness These then lead to requirements for the new system.

→ Models of the new system are used during requirements engineering to help explain the Proposed requirements to other system stakeholders, used to discuss design Proposals & to document the system for implementation.

3- 4 main perspectives in system modeling are:

(A) An external Perspective where you model the context or environment of the system

(B) An interaction Perspective where you model the interactions between a system & it's environment or between components of a system.

(C) A structural Perspective where you model the organization of a system or the structure of the data that is Processed by the system.

(D) A behavioral Perspective where you model the dynamic behavior of the system & how it responds to events

4- Process models ⇒ reveal how the system being developed is used in broader business processes

Context models ⇒ used to illustrate the operational context of a system, they show what lies outside the system boundaries

5- Modeling user interaction is important as it helps to identify user requirements

6- Structural models display the organization of a system in the of the components that make up that system and their relationships. it show the structure of the system design or dynamic models.

7- Real time systems are often event-driven, with minimal data processing, it shows how a system responds to external & internal events, it's based on the assumption that a system has a finite number of states & that events (stimuli) may cause a transition from one state to another.

8- It's an approach to software developments where models rather than programs are the principal outputs of the development process, programs that execute on a hardware / software platform are then generated automatically from the models.

9- It was the precursor of more general model-driven engineering it's a model focused approach to software design & implementation that uses a subset of UML models to describe a system.

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10- Many business systems are data processing systems that are primarily driven by data. They are controlled by the data input to the system with relatively little external event processing, it shows the sequence of actions involved in processing input data & generating an associated output

HCQ:

1- B
2- C
3- B
4- B
5- C
6- B
7- B
8- C
9- B
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